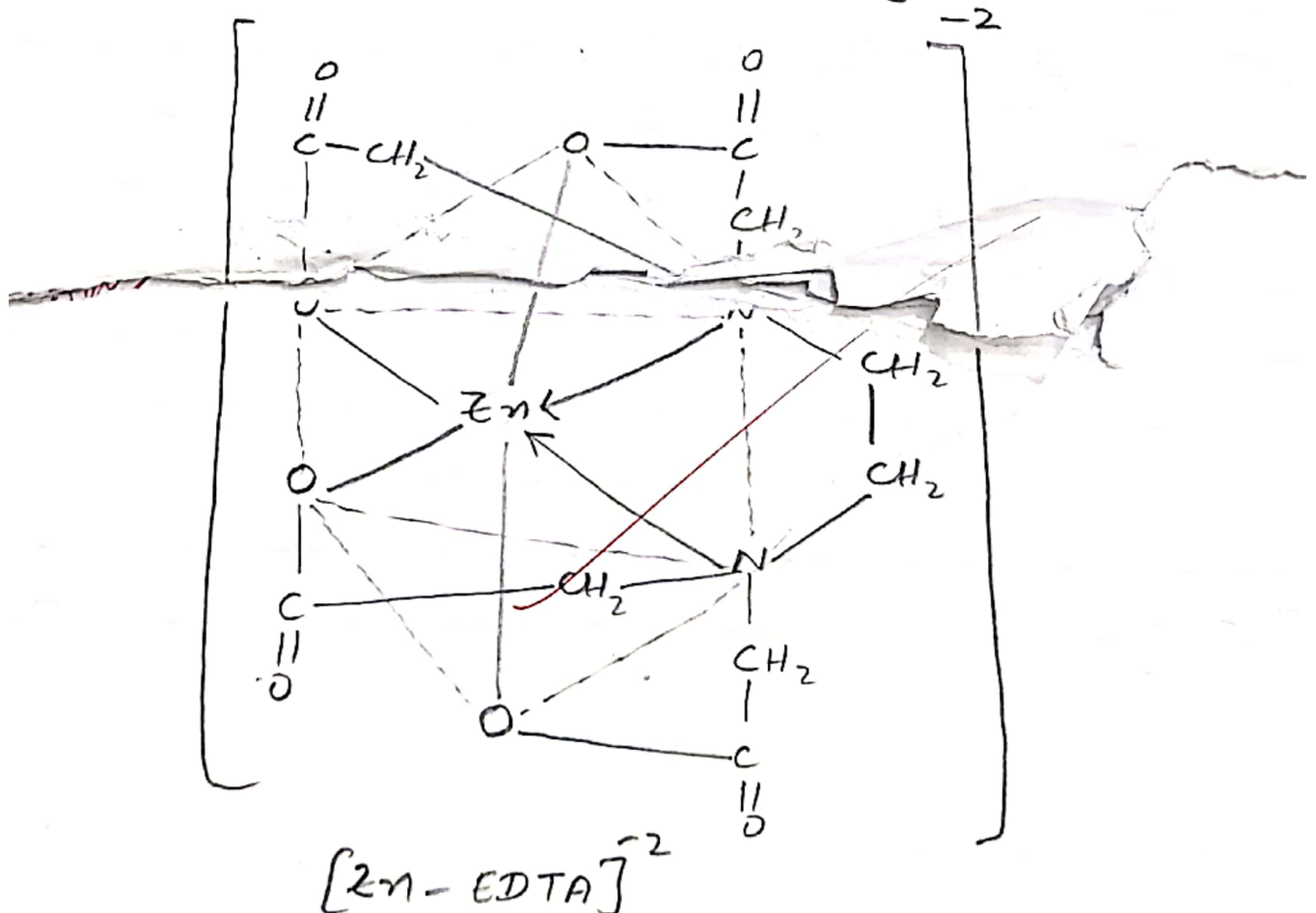
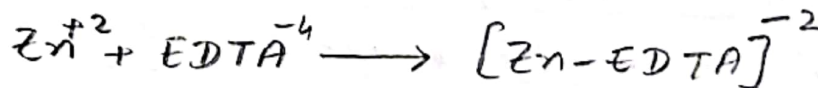
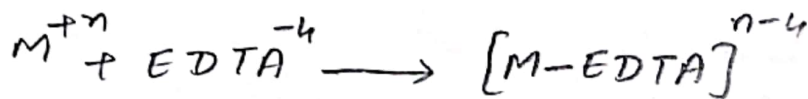
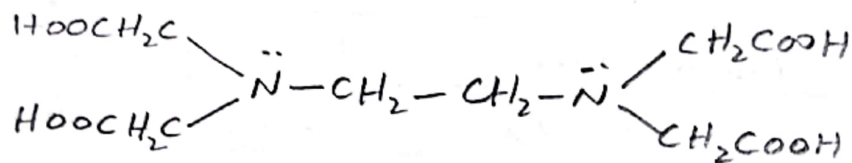


Experiment NO.9

Determine amount/litre of Zn^{+2} and $ZnSO_4 \cdot 7H_2O$ using 0.025M EDTA Solution

EDTA: EDTA stands for Ethylenediamine tetra acetic Acid



Experiment NO. 9
Determine the amount per litre of Zn^{+2} and $ZnSO_4 \cdot 7H_2O$ using 0.025M EDTA solution

principle:

This is direct EDTA titration

Theory:

EDTA titrations are also called complexometry titrations. These are following four types of EDTA titrations

- (i) Direct EDTA titrations
- (ii) Indirect / back EDTA titration
- (iii) Acid-base titration
- (iv) Substitution / replacement EDTA titrations

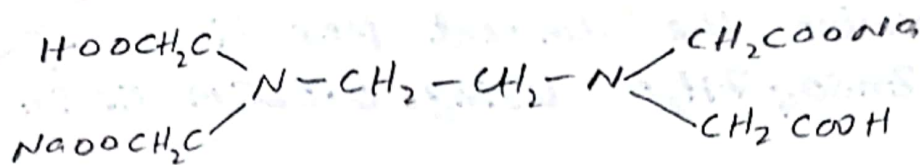
Indicator:

Eriochrome black T (EBT) is used as indicator. It is prepared as following for 1% EBT solution, 0.2 g powder (EBT) + 15 ml triethanolamine + 5 ml ethanol

Triethanol acts as masking agent
Ethanol acts as diluent agent.

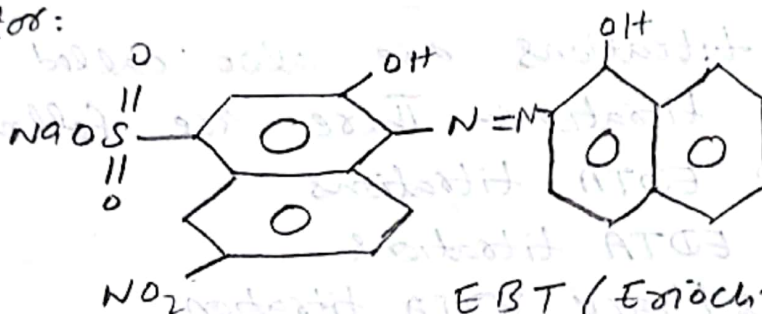
Buffer solution:

Buffer of 10 pH is used to maintain pH of solution. Because metal EDTA complex occurs at pH 10. So buffer of pH 10 is used. Buffer is prepared by taking NH_4Cl 17.5g and liq. NH_3 142 ml and diluting mixture upto 250 ml with distilled water



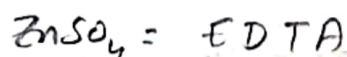
Na_2EDTA
disodium salt of EDTA

Indicator:



EBT (Eriochrome Black T)
Bidentate Ligand

Mole ratio =



End point =

sky blue colour

calculations:

Sr. No	Initial reading (mL)	Final reading (mL)	Volume of EDTA used (mL)
1	0.0	10.1	10.1
2	0.0	10.1	10.1
3	0.0	10.3	10.3
4	0.0	10.4	10.4
5	0.0	10.1	10.1

Concordant reading = 10.1 mL

EDTA stands for ethylenediamine tetra acetic Acid. Its disodium salt is used for increasing solubility tetrasodium salt is not used because it results in hydrolysis of water to a greater extent which increase pH of solution resulting in ppt of metal hydroxide and hinders metal complexation reaction. The other reason is that trans form is more stable than cis form.

uses of Zinc:

Zn is micronutrient. Its atomic number is 30, its atomic mass is 65.4 amu. It is called non-typical transition metal because its d-orbital is completely filled. It still shows complexation reaction. $ZnSO_4$ is used in fertilizers. It is used in forming alloys Zn^{+2} exist in body as central metal.

Procedure:

10 mL sample of $ZnSO_4$ + 2 mL of buffer solution + 2-3 drops of indicator were taken in flask. The colour of indicator was blue. The colour of metal indicator complex was (wine red, purple). Then metal EDTA complex colour was colourless.

Mixture was titrated with EDTA till end point i.e. sky blue colour appeared. Same procedure was repeated to get concordant readings.

Sample	EDTA
$\frac{M_1 V_1}{n_1} = \frac{M_2 V_2}{n_2}$	
$\frac{M_1 \times 10}{1} = \frac{0.025 \times 10.1}{1}$	
$M_1 = 0.025 M$	

$$\text{A/L of } ZnSO_4 \cdot 7H_2O = 0.025 \times 287 = 7.175 \text{ g/L}$$

$$\text{A/L of } Zn^{+2} = 0.025 \times 65.4$$

$$\text{Amount in g/L of } Zn^{+2} = 1.65 \text{ g/L}$$

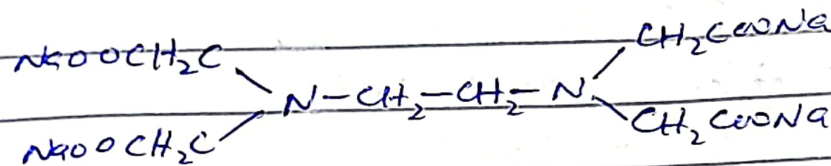
Result:

The given solution containing 7.175 g/L of $ZnSO_4$ and 1.65 g/L of Zn^{+2}

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Why tetra Na is not used?

Because it cause hydrolysis of ions in excess to form OH^- ions which increase pH of solution that form complex with metal hydroxide and hinder metal complexation reaction



Q. why not take cis position?

Because molecule is not stable due to steric hindrance

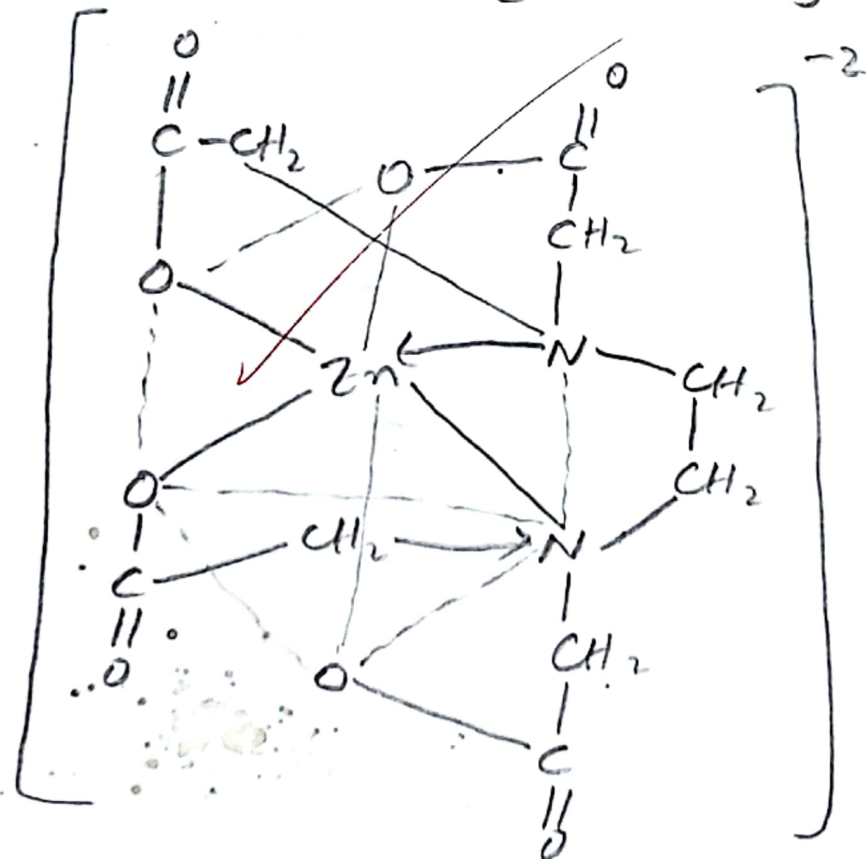
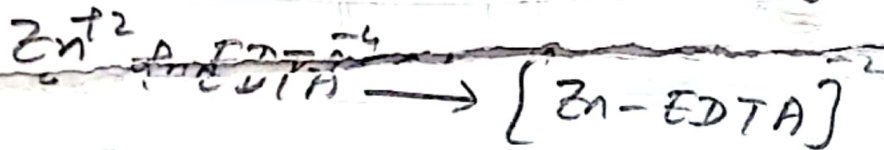
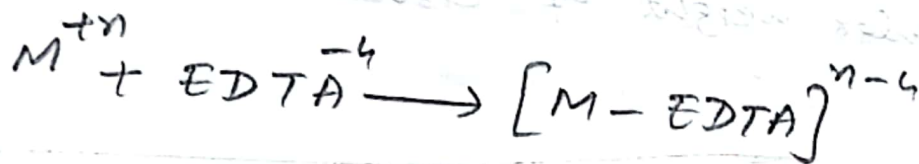
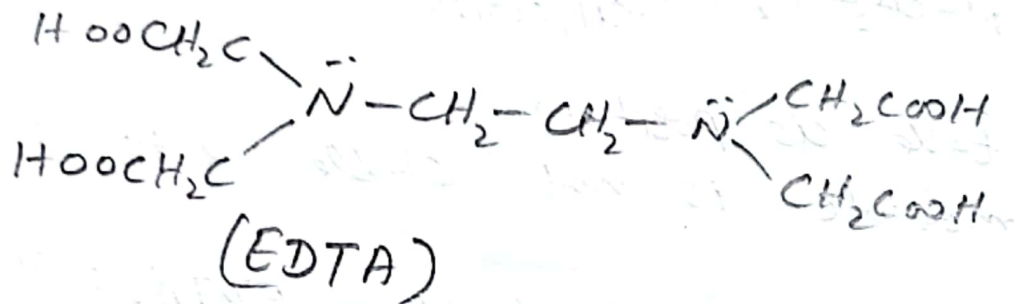
The molecular weight of disodium EDTA salt is 372 g/mol

Experiment No. 10

Standardize given solution of EDTA
using 0.025 M $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$ solution.
Also prepare 0.01 M EDTA solution.

EDTA:

EDTA stands for Ethylenediamine
tetra acetic Acid



Experiment No. 10

Standardize the given solution of EDTA using 0.025M $ZnSO_4 \cdot 7H_2O$ solution. Also prepare 0.01M EDTA solution.

Principle:

This is direct EDTA titration.

Theory:

EDTA titration are also called complexometry titrations. They are of following types

- (i) Direct EDTA titration
- (ii) Indirect or back EDTA titration
- (iii) Alkalimetric/Acid-base
- (iv) Substitutional/replacement titrations.

~~Zn amount can be determined by EDTA titration~~

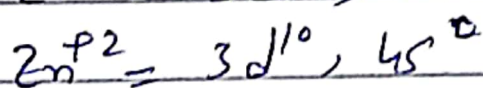
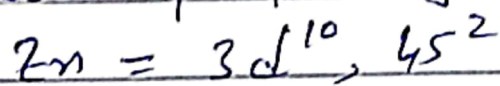
⇒ Gravimetrically using anthranilic acid (amino benzoic Acid)

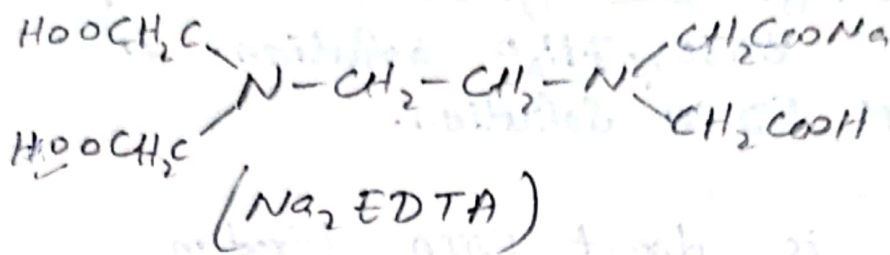
⇒ By atomic absorption method

Zn is non-typical transition metal.

Typical transition metal are those which have d or f orbitals which are in process of filling either in atomic state or ionic state.

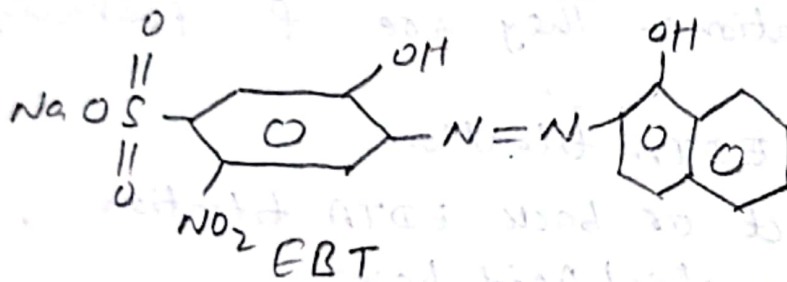
Non typical transition metals have filled d or f orbitals, they are not in process of filling



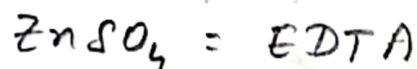


disodium salt of EDTA

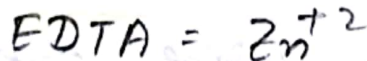
Indicator:



Mole ratio =



$$1 = 1$$



$$1 = 1$$

End Point = Purple/wine red colour

Calculations:

Sr. No	Initial reading	Final reading	Volume used mL
1	0.0	13.0	13.0
2	0.0	13.0	13.0
3	0.0	13.2	13.2
4	0.0	13.1	13.1
5	0.0	13.0	13.0

Zn in both cases have filled d-orbital so it is non-typical transition metal

Indicator:

Eriochrome black T (EBT) is used as an indicator. It is prepared as following

For 1% EBT solution, 0.2g EBT powder + 15ml triethanolamine + 5ml of Ethanol are mixed. Triethanol acts as a masking agent. Ethanol acts as diluent agent

Buffer Solution:

Buffer of pH 10 is used. It is used to maintain pH 10 of solution.

EDTA complexation occurs at pH 10, Buffer is prepared by ~~adding~~ 17.5g NH_4Cl and 142ml liq. NH_3 , diluting the mixture up to 250ml with distilled water.

EDTA stands for ethylenediamine tetraacetic acid. Its disodium salt is used for increasing solubility. Tetraacetic salt is not used because its result in hydrolysis of water to a great extent which increase pH of solution resulting in ppt of metal hydroxide and hinders metal complexation reaction. The other reason is that trans form in disodium salt is more stable than cis form of tetrasodium salt.

Concordant reading = 13.0 mL

EDTA = Zn^{+2}

$$\frac{M_1 V_1}{n_1} = \frac{M_2 V_2}{n_2}$$

$$\frac{M_1 \times 10}{1} = \frac{0.025 \times 13.0}{1}$$

$$M_1 = 0.033M$$

Preparation of 0.01M EDTA Solution:

Given Required

$$C_1 V_1 = C_2 V_2$$

$$0.033 \times V_1 = 0.01 \times 100$$

$$V_1 = 30.3 \text{ mL}$$

Sr. No	Initial reading (mL)	Final reading (mL)	Volume used $ZnSO_4$ mL
1	0.0	4.0	4.0
2	0.0	4.1	4.1
3	0.0	4.0	4.0
4	0.0	4.2	4.2
5	0.0	4.0	4.0

Concordant reading = 4.0 mL

Result:

The standardized solution of 0.01M EDTA was prepared.

uses of Zn:

Zn^{0} is a micronutrient. Its atomic number is 30 while its atomic mass is 65.4. It is called non-typical transition metal but it still shows complexation reaction. $ZnSO_4$ is used in fertilization. It is used in form of alloy.

Procedure:

10 mL sample of $ZnSO_4$, 2 mL buffer solution, 2-3 drops of indicator were taken in a flask. The colour of indicator was blue. The colour of metal-indicator complex was wine red (purple). Then metal EDTA complex was colourless. Mixture was titrated against EDTA till end point sky blue colour was appeared.

Same procedure was repeated to get three concordant readings.

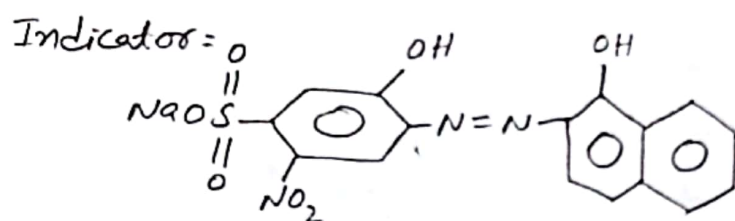
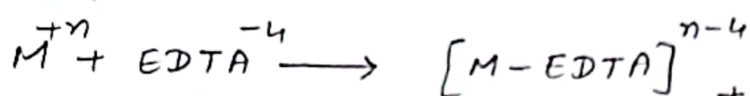
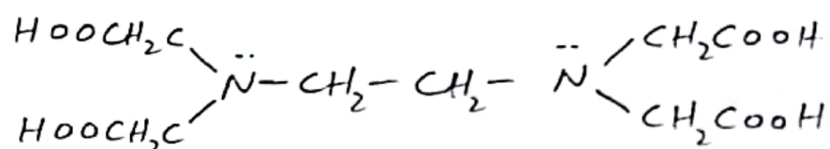
uses of EDTA

- (i) It is used as food preservatives
- (ii) It removes undesirable metal from body
- (iii) It is used to remove the effect of heart attack
- (iv) It is used to remove angina and chest pain
- (v) It is used to reduce blood pressure in about 60% of high blood pressure patients.

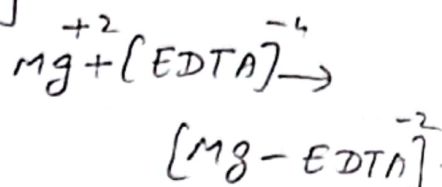
Experiment No. 12

Determine A/L of Magnesium ions and Magnesium Sulphate in given Sample Solution using 0.025M EDTA solution.

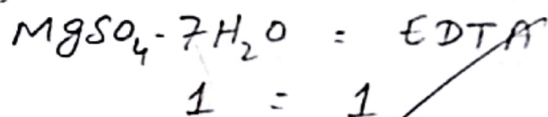
EDTA: EDTA stands for Ethylenediamine tetra acetic Acid



EBT



Mole ratio =



End Point:

Sky blue colour

calculations:

Sr. No	Initial reading	Final reading	Volume used (mL) EDTA
1	0.0	11.3	11.3
2	0.0	11.2	11.2
3	0.0	11.4	11.4
4	0.0	11.3	11.3
5	0.0	11.3	11.3

Experiment No. 12

Determine Amount per litre of Magnesium ions and Magnesium Sulphate in given Sample Solution using 0.025M EDTA Solution.

Principle:

Theory:

EDTA titrations are also called Complexometry titrations. They are of following types

- (i) Direct EDTA titration
- (ii) Indirect or back EDTA titration
- (iii) Alkalimetric / Acid-base titration
- (iv) Substitutional / replacement titrations

Indicator:

Eriochrome black T is used as indicator. It is prepared as following for 1% EBT solution, 0.2g powder + 15mL triethanolamine + 5mL Ethanol

Triethanol acts as masking agent.
Ethanol acts as diluent agent.

Buffer Solution:

Buffer of pH 10 is used to maintain pH of solution. Because metal EDTA complexation occurs at pH 10. So buffer of pH 10 is used. Buffer is prepared by taking 17.5g NH_4Cl + 142 mL liquid NH_3 + dilute it upto 250 mL with distilled water

Concordant Reading = 11.3 mL

Calculations:

Sample = EDTA

$$\frac{M_1 V_1}{n_1} = \frac{M_2 V_2}{n_2}$$

$$\frac{M_1 \times 10.0}{1} = \frac{0.025 \times 11.3}{1}$$

$$M_1 = 0.028 \times 1$$

$$M_1 = 0.028$$

$$\begin{aligned} \text{A/L of } \text{Mg}^{+2} \text{ ions} &= 0.028 \times 24 \\ &= 0.678911 \end{aligned}$$

$$\begin{aligned} \text{A/L of } \text{MgSO}_4 \cdot 7\text{H}_2\text{O} &= \text{Molarity} \times 242 \\ &= 0.028 \times 242 \\ &= 6.77916 \end{aligned}$$

Result:

The given Solution contains 0.6789/L of Mg^{2+} and MgSO_4 6.779/L

ACH
4-11-13

EDTA stands for Ethylene diamine tetra acetic Acid. Its disodium salt is used for increasing solubility tetrasodium salt is not used because its results in hydrolysis of water to a great extent which increase pH of solution.

Uses of Magnesium:

Magnesium is an alkaline earth metal. It is used in preparation of fire explosive material E.g Magnesium ribbon. Magnesium oxide is used as suspension known as milk of magnesia for treatment of disorders.

The presence of magnesium in water cause water hardness.

Two types of hardness.

(i) Temporary hardness

(ii) Permanent hardness

Temporary cause due to bicarbonates of calcium and magnesium. Permanent cause by chlorides and Sulphate. For treatment of hard water heating calcium oxide and some ion exchange resins are used E.g Sodium zeolite is used presence of hardness water produce scaling problem and also create problems in washing bicurde. Magnesium also important component of chlorophyll.

procedure:

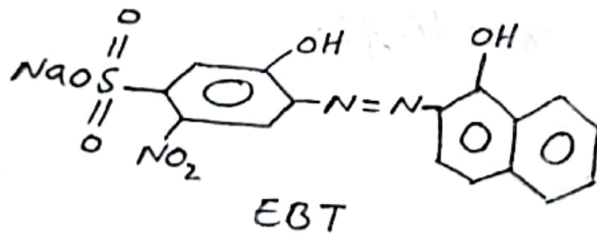
10ml of sample soln. + 2ml of buffer soln. + 2-3 drops of EBT was taken in flask. The solution is wine red colour titrate it against EDTA until end point. Repeat this experiment to get three concordant readings.

Experiment NO-13

Determine Amount per Litre of Nickel ion Ni^{2+} and Nickel Sulphate $\text{NiSO}_4 \cdot 6\text{H}_2\text{O}$ in given Sample Solution using 0.025M EDTA Solution and 0.025M $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$

principle: Indirect EDTA titration

Indicator:



Mole ratio:

EDTA : Zinc Sulphate
1 : 1

EDTA : Nickel ion
1 : 1

End point:

Sky blue colour

calculations:

Sl. No.	Initial reading	Final reading	volume used ZnSO_4 (mL)
1	0.0	9.5	9.5
2	0.0	9.4	9.4
3	0.0	9.5	9.5
4	0.0	9.3	9.3
5	0.0	9.5	9.5

Concordant reading = 9.5 mL

Experiment NO-13

Determine Amount per Litre of Nickel ion Ni^{2+} and Nickel Sulphate $\text{NiSO}_4 \cdot 6\text{H}_2\text{O}$ in given Sample Solution using 0.025M EDTA Solution and 0.025M $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$

Theory:

This is indirect EDTA titration. EDTA titration are also called Complexometry titrations. They are of following types

- (i) Direct EDTA titration
- (ii) Indirect or back EDTA titration
- (iii) Alkalimetric/ Acid-base
- (iv) Substitutional / replacement titration

Indirect EDTA titration are used when the complex formation of metal ion with EDTA is slow and reversible.

Indicators:

Eriochrome black T is used as indicator.

It is prepared as following for 1% EBT Solution,
0.2g powder + 15ml triethanolamine + 5ml Ethanol

Triethanol acts as making agent.

Ethanol acts as diluent agent.

Buffer solution:

Buffer of pH 10 is used to maintain pH of solution. Because metal EDTA Complexation occurs at pH 10. So buffer of pH 10 is used.

Buffer is prepared by taking 17.5g NH_4Cl + 142ml liquid NH_3 + dilute it upto 250ml with distilled water.

calculations:

$Ni^{+2} = EDTA$

$$\frac{M_1 V_1}{n_1} = \frac{M_2 V_2}{n_2}$$

$$\frac{0.025 \times 9.5}{1} = \frac{0.025 \times V_2}{1}$$

$$V_2 = 9.5 \text{ mL}$$

Volume of EDTA that react with Nickel ion Ni^{+2}

$$V_4 = 20 - V_2$$

$$= 20 - 9.5 = 10.5$$

$Ni^{+2} = EDTA$

$$\frac{M_3 V_3}{n_3} = \frac{M_4 V_4}{n_4}$$

$$\frac{M_3 \times 10.0}{1} = \frac{0.025 \times 10.5}{1}$$

$$M_3 = 0.026 \text{ M}$$

A/L of Nickel ion $Ni^{+2} = M_3 \times 5.9$

$$= 0.026 \times 5.9$$

$$= 1.538 \text{ g/L}$$

A/L of Nickel sulphate = $M_3 \times 263$

$$= 0.026 \times 263$$

$$= 6.838 \text{ g/L}$$

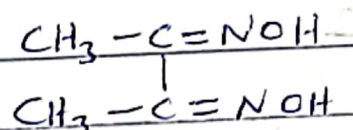
Result: The given solution contains 1.538 g/L of Nickel ions and 6.838 g/L of Nickel sulphate.

EDTA:

EDTA stands for Ethylene diamine tetra Acetic acid. Its disodium salt is used for increasing solubility. Tetrasodium salt is not used because it results in hydrolysis of water to a great extent which increases pH of solution.

Nickel theory:

Nickel is the transition element with atomic number 28 and it is a micronutrient for plants. It is used in many industries like electroplating and fertilizers. It is also used for hydrogenation of vegetable oil into vegetable ghee. Higher concentration of nickel in water produces skin disease. It can also be determined gravimetrically by using DMC.



procedure:

10 mL Sample (Nickel ion) + 20 mL EDTA + 2 mL Buffer solution + 2, 3 drops Indicator EBT. A wine red colour is produced. Then titrate it against Zinc Sulphate. When sky blue colour is produced. Similarly take three more readings until end point.

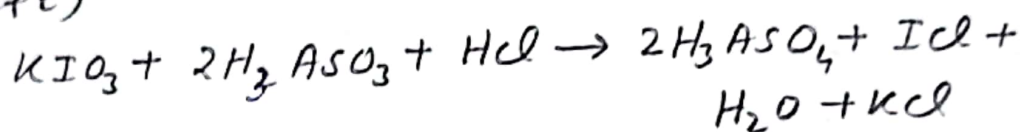
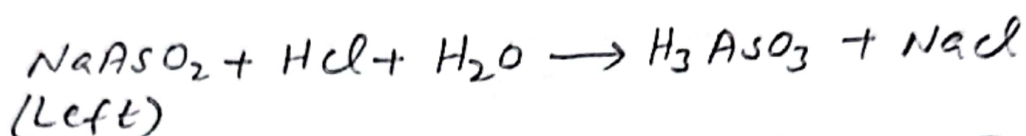
Experiment No. 15

Determine the A/L of NaOCl in the given sample using 0.025 KIO₃ and 0.05M NaAsO₂

Indicator:

chloroform / CCl₄

Chemical Equation:



Mole ratio:



$$1 = 1$$



$$1 = 2$$

End point:

organic layer — colourless

Aq. layer — lemon yellow

Sr. No	Initial reading	Final reading	Volume used (mL)
1	0.0	9.8	9.8
2	9.8	19.8	9.8
3	0.0	9.8	9.8
4	9.8	19.9	9.9
5	0.0	19.8	9.8

Concordant volume = 9.8 mL

Experiment NO. 15

Determine the A/L of NaOCl in given sample solution using 0.025 KIO₃ and 0.5M Na₂SO₃.

Principle:

This is an iodimetric direct titration

Theory:

Iodine titration are of two types

(i) By indirect method (Iodometric titration)

(ii) By direct method (Iodimetric titration)

Indicator:

According to the environment of given mixture two types of indicators can be used

(i) For less acidic medium starch is used. In this case end point is disappearance of blue colour

(ii) For more acidic medium CHCl₃, CCl₄ and CS₂ is used. In this case organic layer should be transparent in lemon yellow colour of remaining mixture.

Liquid Bleach:

(i) Its formula is NaOCl

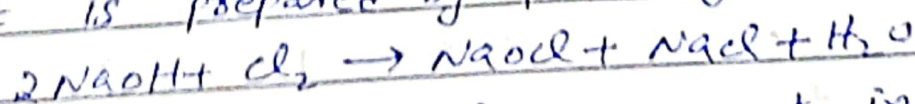
(ii) Colourless to yellowish colour solution

(iii) Atomic mass 74.4

(iv) It has peculiar Cl pungent smell

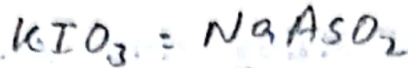
(v) Density = 1.2 g/L

(vi) It is prepared by following process



(vii) It is used as cleansing agent in lime industry, food industry, chemical industry for cleansing of instrument.

Calculation:



$$\frac{M_1 V_1}{n_1} = \frac{M_2 V_2}{n_2}$$

$$\frac{0.025 \times 9.1}{1} = \frac{0.05 \times V_2}{2}$$

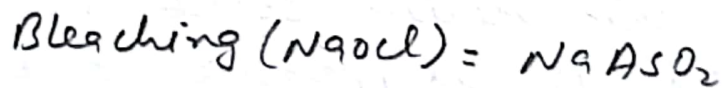
$$V_2 = 9.8 \text{ mL}$$

Volume of NaAsO_2 reacted with

$$\text{Sample} = V_4 = 20 - V_2$$

$$= 20 - 9.8$$

$$= 10.2 \text{ mL}$$



$$\frac{M_3 V_3}{n_3} = \frac{M_4 V_4}{n_4}$$

$$\frac{M_3 \times 10.0}{1} = \frac{0.05 \times 10.2}{1}$$

$$M_3 = 0.051 \text{ M}$$

$$\text{Al of NaOCl} = M_3 \times \text{Mol. wt}$$

$$= 0.051 \times 74.4$$

$$= 3.79 \text{ g/L}$$

Result:

Am/L of NaOCl in given sample was 3.79 g/L

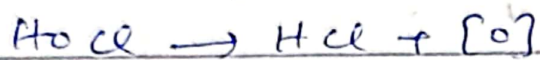
Adh

It is also used as bleaching agent.

(ix) It is used to detoxified CN both in metal industry, by oxidizing it in CO_2 .

(x) It is used to prevent bacterial growth.

(xi) NaOCl disinfect water



HCl provide acidic media

(xii) It is dangerous and corrosive substance, as it oxidized other substance by oxidation due to $[\text{O}]$

procedure:

10mL Sample NaOCl + 28mL CHCl_3 (as indicator) + 20mL NaAsO_2 + 4-5mL HCl were taken in an iodine flask. It was then subjected to titration against KIO_3 till end point which is yellow colour of aqueous layer with a colourless layer of organic liquid in it. The procedure was repeated to get concordant reading.